DATA MODELING AND VISUALIZATION (DMV) PROJECT REPORT ON

“ASD Insight - Eye Image Research Prototype”

SUBMITTED TOWARDS THE
PARTIAL FULFILLMENT OF THE REQUIREMENTS

OF **BACHELOR OF ENGINEERING (BE AI & DS)**

BY

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Under The Guidance of

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“Excellence in the field of AI & DS”

Department of Artificial Intelligence & Data Science

“Techno-Social Excellence”

Marathwada Mitramandal’s Institute of Technology

Lohgaon, Pune- 411 047
A. Y (2025-26)



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CERTIFICATE

This is to certify that the Project Entitled

"ASD Insight - Eye Image Research Prototype"

Submitted by

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are bonafide work carried out by students under the supervision of **Dr. Shrikant Dhamdhere** and it is submitted towards the partial fulfillment of the requirement of Bachelor of Engineering (BE AI & DS) Software laboratory II

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ABSTRACT

ASD Insight is an experimental research application that explores image-based pattern classification using eye-region images labeled as ASD and non-ASD. The project provides a browser-based interface built with React, Vite, and Tailwind CSS, connected to a Python FastAPI backend. A ResNet18-based image classification model is trained with PyTorch using labeled images, with training and validation splits to support initial evaluation. Users can upload an eye-region image and view the model's predicted class and score. This system is intended only for educational and experimental research. Eye images alone cannot establish whether a person has autism, and the output must not be used for screening, diagnosis, or clinical decisions. Results require careful validation on independent, representative datasets.

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INTRODUCTION

Artificial intelligence and computer vision are increasingly used to study image patterns in research. Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition whose identification requires comprehensive professional assessment; it cannot be reliably diagnosed from an eye-region photograph. This project therefore frames image classification as an experimental pattern-recognition exercise, not as an ASD diagnostic tool. ASD Insight allows a user to upload an eye-region image, preprocesses the image, and passes it to a trained deep-learning classifier that predicts one of the dataset labels, ASD or non-ASD. The interface is implemented using React, Vite and Tailwind CSS, while the backend uses Python FastAPI to validate uploads and serve model inference.

OBJECTIVES

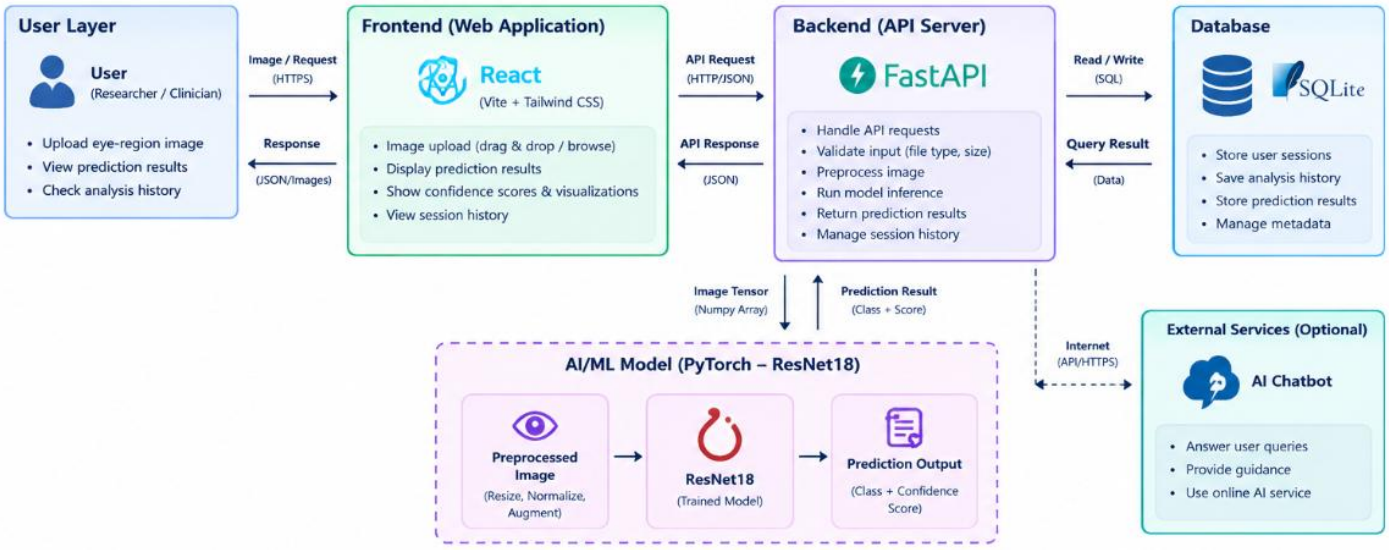
- To develop an experimental image-classification application for labeled eye-region images.
- To train a ResNet18-based model using PyTorch for the dataset's two labels: ASD and non-ASD.
- To build a responsive user interface using React, Vite and Tailwind CSS.
- To implement a Python FastAPI backend for image upload and model inference.
- To apply consistent image resizing and normalization before inference.
- To report model outputs transparently and evaluate performance with suitable metrics.
- To document limitations, dataset bias, privacy concerns and the need for independent validation.
- To make clear that predictions are experimental and are not a medical diagnosis.

PROBLEM STATEMENT

Title: ASD Insight - Eye Image Research Prototype

Problem Statement: Labeled eye-region image datasets can be used to explore computer-vision methods for pattern classification, but preparing images, running inference and presenting results consistently can be inconvenient without an integrated application. This project develops a research prototype with a web interface and API backend for experimenting with ASD/non-ASD dataset labels. The prototype does not determine whether a person has ASD and must not be used for medical screening or diagnosis.

SYSTEM ARCHITECTURE



The user selects an eye-region image in the React frontend. The frontend sends the file to the FastAPI backend over an HTTP request. The backend checks that the upload is an acceptable image, decodes it, resizes and normalizes it using the model's expected preprocessing, and passes the resulting tensor to the trained PyTorch model. The model returns class scores, which the API serializes as a response for the frontend to display. The result is explicitly labeled experimental. A model score is not a clinical probability and does not establish an individual's ASD status.

PROPOSED SYSTEM

The proposed system integrates an image-upload interface with a machine-learning inference service. The frontend is developed using React, Vite and Tailwind CSS. The backend is implemented in Python using FastAPI, and the image classifier uses a ResNet18 architecture with PyTorch. The dataset contains two folders/classes, ASD and non-ASD. Training, validation and testing must be separated carefully to avoid data leakage. The prototype presents predictions for research demonstrations only and should not make diagnostic claims.

HARDWARE REQUIREMENT

- Processor: Intel i3/i5 or equivalent and above.
- RAM: 8 GB recommended; 4 GB may be restrictive for training.
- Storage: Sufficient free space for source code, dependencies and dataset.
- GPU: Optional; training can run on CPU but may be slower.
- Input/output: Keyboard, mouse and display.

SOFTWARE REQUIREMENT

• Operating System: Windows, Linux or macOS.

• Frontend: JavaScript, React, Vite and Tailwind CSS.

• Backend: Python and FastAPI; Uvicorn ASGI server.

• Machine learning: PyTorch and Torchvision (ResNet18), Pillow and scikit-learn for evaluation where applicable.

• Development tools: Visual Studio Code and a modern web browser.

• Internet connection: Required for initial dependency installation and any pretrained-weight download.

IMPLEMENTATION

1. Dataset organization: Place labeled eye-region images in separate ASD and non-ASD folders. Keep dataset permissions and privacy requirements in mind.

2. Preprocessing: Decode each image, convert to RGB, resize to the model's input size and normalize using the preprocessing expected by the model.

3. Data splitting: Create training and validation sets, and preferably keep an independent test set. If images come from the same person, split by person rather than by image to reduce leakage.

4. Model development: Use a ResNet18 image classifier implemented in PyTorch, with the final classification layer adapted for two dataset labels.

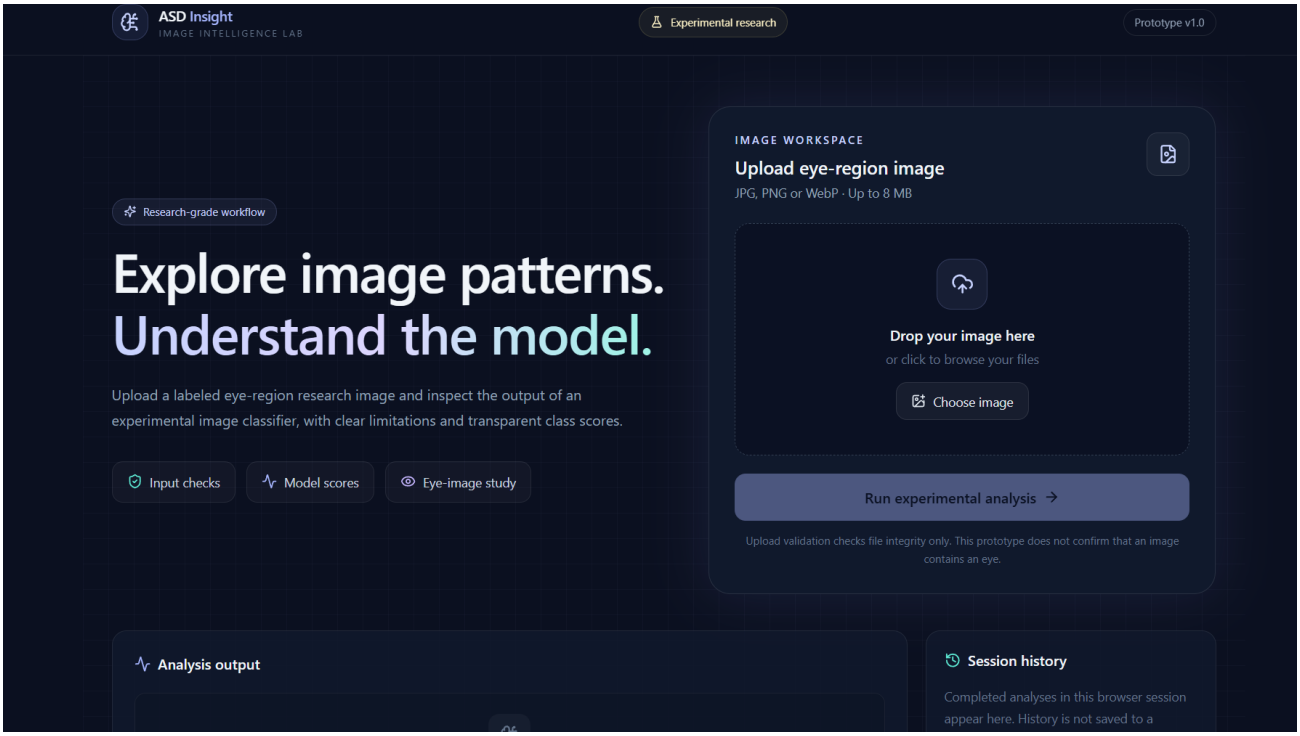
5. Training and evaluation: Train using the training split, monitor validation loss and relevant metrics, and report confusion matrix, precision, recall, F1-score and sensitivity/specificity where appropriate. Do not claim clinical accuracy without rigorous independent evaluation.

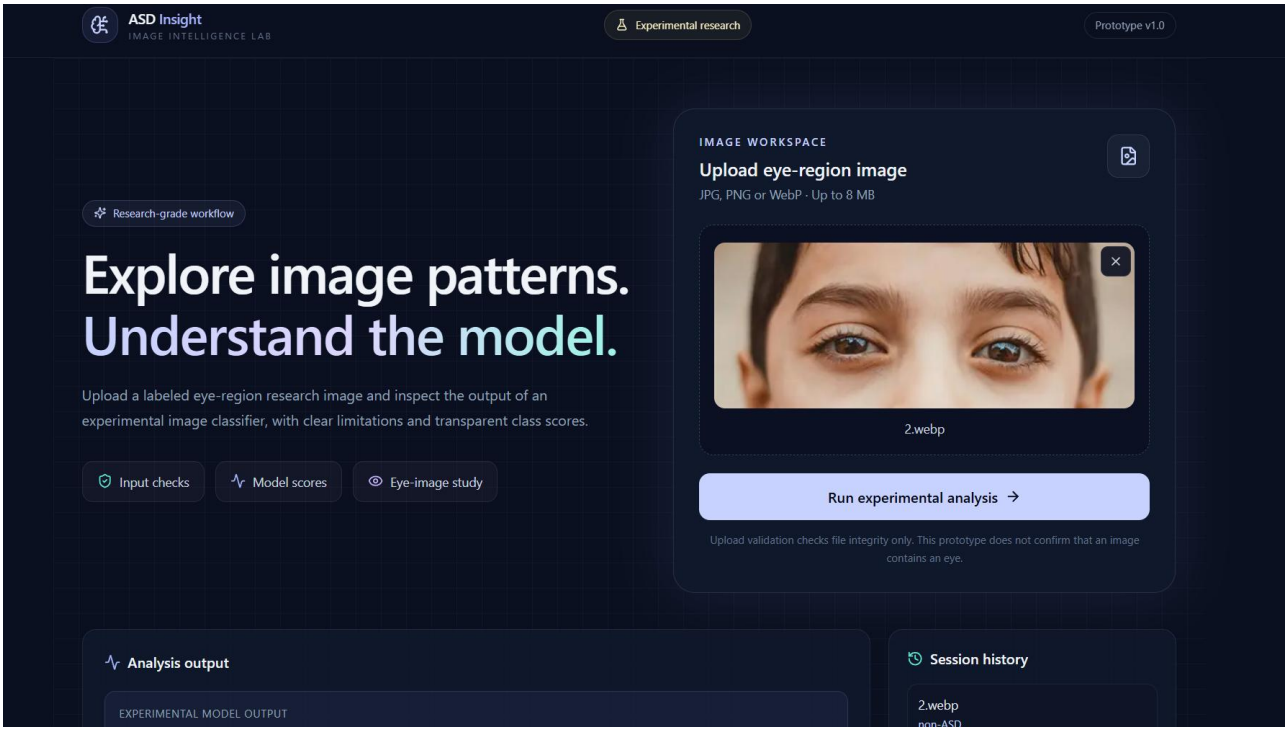
6. Backend: FastAPI receives uploaded images, validates file decoding and dimensions, applies preprocessing and returns the model's output as JSON.

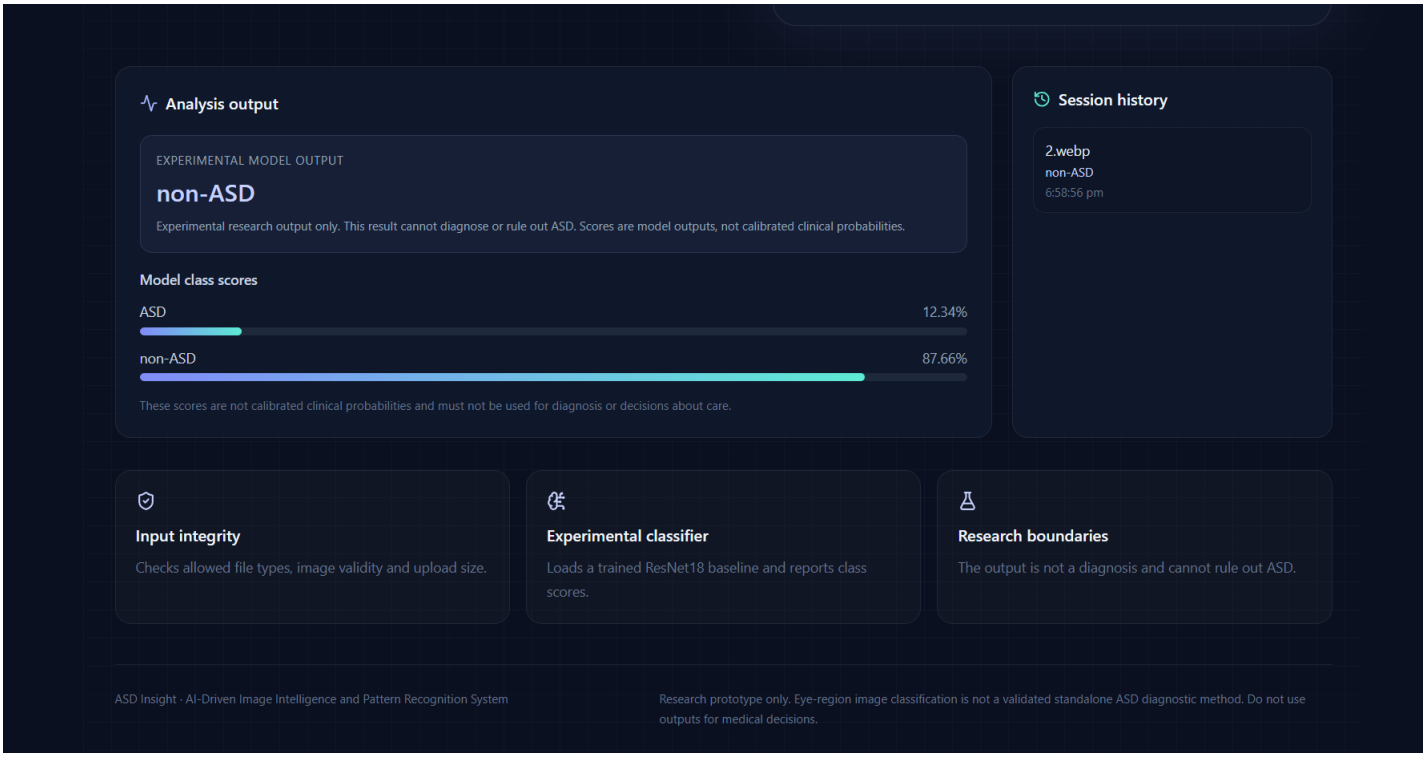
7. Frontend: React/Vite/Tailwind provides image selection, preview, request status and a result panel. It should show an experimental-use disclaimer alongside every prediction.

8. Testing: Test valid images, invalid files, oversized uploads, missing model files, API errors and frontend connection failures. Confirm that the app never describes the output as a diagnosis.

SNAPSHOTS OF SYSTEM






Analysis output
EXPERIMENTAL MODEL OUTPUT
non-ASD
Experimental research output only. This result cannot diagnose or rule out ASD. Scores are model outputs, not calibrated clinical probabilities.
Model class scores
ASD
12.34%
non-ASD
87.66%
These scores are not calibrated clinical probabilities and must not be used for diagnosis or decisions about care.
Session history
2.webp
non-ASD
6:58:56 pm
Input integrity
Checks allowed file types, image validity and upload size.
Experimental classifier
Loads a trained ResNet18 baseline and reports class scores.
Research boundaries
The output is not a diagnosis and cannot rule out ASD.
ASD Insight · AI-Driven Image Intelligence and Pattern Recognition System
Research prototype only. Eye-region image classification is not a validated standalone ASD diagnostic method. Do not use outputs for medical decisions.

ADVANTAGES

- Integrates image upload, inference and result display in one application.
- Provides a practical educational example of a computer-vision workflow.
- Separates frontend and backend responsibilities.
- Uses a widely used transfer-learning architecture as a baseline.
- Can be extended for research evaluation, reproducibility and model comparison.

DISADVANTAGES

- Dataset quality, label reliability and representativeness strongly affect results.
- The model may learn unwanted dataset or image-acquisition artifacts.
- CPU training can be slow.
- A high score on a small or non-independent test set may not generalize.
- Eye-region images alone cannot diagnose ASD; the prototype is not a clinical tool.
- Privacy, consent, demographic bias and secure image handling require careful attention.

CONCLUSION

ASD Insight demonstrates an end-to-end experimental image-classification workflow by combining a React, Vite and Tailwind CSS interface with a FastAPI backend and a PyTorch ResNet18 model. It allows users to submit eye-region images and inspect the model's predicted dataset label. The project's value is as a learning and research prototype for image processing, model serving and evaluation. Its predictions do not establish whether an individual has autism and must not be used for diagnosis, screening or clinical decisions. Further work requires transparent evaluation, independent testing, bias analysis and appropriate ethical review.

FUTURE SCOPE

• Evaluate multiple architectures and compare them using a fixed independent test set.

• Add reproducible reports for confusion matrix, precision, recall, F1-score and calibration.

• Improve dataset documentation, consent practices and privacy-preserving storage.

• Test for demographic and acquisition-related bias and potential data leakage.

• Add explainability tools for model-behavior research, with careful interpretation.

• Improve upload validation, accessibility, logging and deployment security.

• Any future clinical application would require appropriate clinical evidence, expert oversight and regulatory review.

REFERENCES

• PyTorch Documentation: https://pytorch.org/docs/stable/index.html

• Torchvision Models Documentation: https://pytorch.org/vision/stable/models.html

• FastAPI Documentation: https://fastapi.tiangolo.com/

• React Documentation: https://react.dev/

• Vite Documentation: https://vite.dev/

• Tailwind CSS Documentation: https://tailwindcss.com/docs

• Kaggle dataset supplied for this project: https://www.kaggle.com/datasets/sha0401/autismdataset

• Project repository: https://github.com/dikshasolase/ASD---Insight-Eye-Image-Research